CrisisLens

NLP for a Safer, More Informed Tomorrow

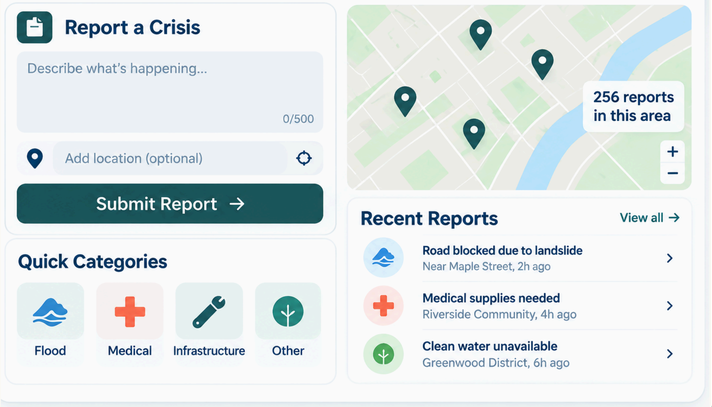
A web-based platform for reporting, classifying and retrieving relevant crisis information.

01 Problem & Need

- Crisis-related information is often fragmented, unstructured and difficult to browse quickly.
- Urgent needs, damage reports and relevant local updates can be easily overlooked.
- People need a clearer way to organise, search and access relevant crisis information.

02 Platform Preview

A simple and intuitive interface for real-world impact.



03 End-To-End Workflow

From a crisis report to actionable information.

1 **REPORT**
Submit crisis text + location

↓

2 **CLASSIFY**
NLP Classification
(category + confidence)
Traditional ↔ Transformer

→ **Low confidence?**
-User confirms
-corrects category

↓

3 **ORGANISE**
Save to Crisis Hub
board • time • location

↓

4 **RETRIEVE**
Search & related reports
Lexical ↔ Semantic retrieval

↓

5 **ACT**
Actionable information for response, support & discovery

04 Why CrisisLens?

Direct Reporting
Users can submit crisis information directly.

NLP Classification
Reports are automatically organised into relevant crisis categories.

Relevant Retrieval
Users can search and discover relevant crisis reports.

One hub for reporting, organising and finding crisis information.

05 Why Is It Different?

Beyond static classification: an integrated, user-guided crisis information workflow.

END-TO-END
Integrated Crisis Workflow
Reporting, classification and retrieval are connected in one workflow.

SEMANTIC
Action-Oriented Retrieval
Semantic search connects users with relevant crisis reports beyond exact keyword matching.

HUMAN-IN-THE-LOOP
User-Guided Classification
Users can confirm or correct low-confidence predictions.

06 Why Does It Matter?

Better information. Faster response.

CrisisLens helps users find and organise relevant crisis information more efficiently, supporting faster and more informed responses.

Faster Discovery · Clearer Organisation · Better Support

07 Use Cases

Affected Communities
- Report local needs
- Find relevant crisis information
- Stay informed

Responders & Volunteers
- Identify relevant incidents
- Locate where support is needed
- Coordinate response efforts

Donors & Supporters
- Discover reported needs
- Track evolving situations
- Find opportunities to support

08 Technical Plan & Evaluation

Data
- HumAID: 76,484 posts
- 19 events · 10 labels
- Official train / validation / test split

Classification
- Traditional ML baseline
- Transformer model
- Multi-class classification
- Confidence estimation

RETRIEVAL
- Lexical retrieval baseline
- Semantic retrieval
- Relevant report ranking
- Location / time / category filters

EVALUATION
- Classification: Macro-F1
- Retrieval: Precision@K, Recall@K
- Error analysis
- User study & task completion

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